

TAMIL NADU STATE BOARD - STANDARD 9 MATHEMATICS

CHAPTER 3: ALGEBRA

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Mathematics
Chapter Name:	Algebra

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

UNIT ROADMAP: ALGEBRAIC IDENTITIES MAPPING



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Polynomial	An algebraic expression consisting of variables and constants, using operations of addition, subtraction, multiplication, and non-negative integer exponents.
Degree of Polynomial	The highest power of the variable in the polynomial.
Remainder Theorem	If a polynomial $p(x)$ is divided by $(x - a)$, the remainder is $p(a)$.
Factor Theorem	If $p(a) = 0$ for a polynomial $p(x)$, then $(x - a)$ is a factor of $p(x)$.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

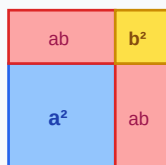
GLOSSARY: ALGEBRAIC TERM HIERARCHY



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: ALGEBRAIC IDENTITY PARTITION SQUARE



Essential Formulas, Laws & Identities:

- Identity: $(x + y)^2 = x^2 + 2xy + y^2$
- Identity: $(x - y)^2 = x^2 - 2xy + y^2$
- Identity: $(x + y)(x - y) = x^2 - y^2$
- Cubic: $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$
- Cubic: $(x - y)^3 = x^3 - y^3 - 3xy(x - y)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

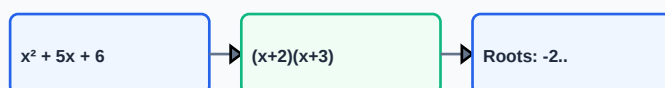
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Factorise $x^2 - 5x + 6$.**Step-by-step Solution:**

Find two numbers whose product is 6 and sum is -5. They are -2 and -3. $x^2 - 2x - 3x + 6 = x(x - 2) - 3(x - 2) = (x - 2)(x - 3)$.

Example 2: Find the remainder when $p(x) = x^3 - 3x^2 + 4x - 5$ is divided by $(x - 2)$.**Step-by-step Solution:**

By Remainder Theorem, remainder is $p(2)$: $p(2) = (2)^3 - 3(2)^2 + 4(2) - 5 = 8 - 12 + 8 - 5 = -1$.

WORKED: FACTORING QUADRATIC TRINOMIAL

HOTS (Higher Order Thinking Skills) Challenge

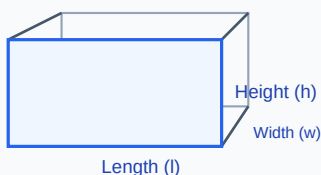
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Find the values of a and b if the polynomial $x^3 - ax^2 + bx - 6$ is exactly divisible by $(x - 1)$ and $(x - 2)$.

Analytical Solution Strategy:

Since it is divisible, $p(1) = 0$ and $p(2) = 0$. $p(1) = 1 - a + b - 6 = 0 \Rightarrow a - b = -5$ (Eq 1) $p(2) = 8 - 4a + 2b - 6 = 0 \Rightarrow 2a - b = 2$ (Eq 2) Subtracting Eq 1 from Eq 2: $a = 6$. Substituting in Eq 1: $b = 11$.

HOTS: ALGEBRAIC IDENTITY IN 3D CUBOID



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Divide $x^3 - 3x^2 + 5x - 3$ by $x - 1$ using synthetic division.

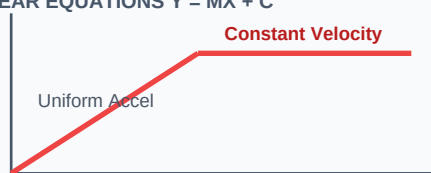
Q2 (2-Mark): Factorise: $2x^2 + 7x + 6$.

Q3 (2-Mark): Expand: $(2a + 3b)^3$.

Q4 (5-Mark): Find the degree of the polynomial: $4x^5 - 3x^2 + 2x - 1$.

Q5 (5-Mark): Solve the linear equation: $3x - 5 = 2x + 7$.

PRACTICE: PLOTTING LINEAR EQUATIONS $Y = MX + C$



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Computer Graphics

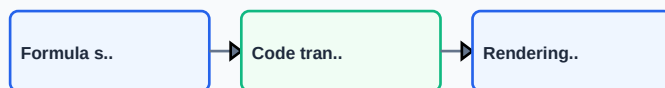
Polynomials model smooth Bézier curves used in digital design, vector drawing, and engineering aerodynamic designs.

Application Case: Economics

Linear equations determine business break-even points, correlating cost and revenue functions to compute profit margins.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: ALGEBRAIC FORMULAS IN GRAPHICS



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: ALGEBRA CONCEPTS FLOW

